Rational and irrational Turán densities: New results for 5-uniform hypergraphs via Lagrangian densities

Jianfeng Hou [*1], Caiyun Hu [†1], and Biao Wu [‡2]

[1]Center for Discrete Mathematics, Fuzhou University, Fujian, 350003, China
[2]School of Mathematics and Statistics, Hunan Normal University, Hunan, 410081, China

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Abstract

For $r \geq 3$, Baber and Talbot asked whether there exist single r-uniform hypergraphs with irrational Turán densities. While such hypergraphs were previously known only for $r = 3$ and $r = 4$, we provide the first construction in the case $r = 5$ by studying the Lagrangian density of the 5-uniform matching $M_4^{(5)}$ of size 4. Precisely, we prove that the Lagrangian density of $M_4^{(5)}$ is an irrational number. As a consequence, a classical result of Sidorenko implies that the Turán density of its extension is also irrational. We also show that the Lagrangian density of a 5-uniform matching of size 3 is a rational number, highlighting a distinction in the densities of these matching structures.

Keywords: Turán number, Lagrangian density, matching, left-compressed

1 Introduction

1.1 Turán densities

For $r \geq 2$ an r-uniform hypergraph (henceforth r-graph) $\mathcal{H}$ is a collection of r-subsets of some finite set V. We call V the **vertex set** of $\mathcal{H}$ and denote it by $V(\mathcal{H})$. We identify a hypergraph $\mathcal{H}$ with its set of edges, and hence, $|\mathcal{H}|$ represents the number of edges in $\mathcal{H}$.

Given a family $\mathcal{F}$ of r-graphs we say $\mathcal{H}$ is $\mathcal{F}$-free if it does not contain any member of $\mathcal{F}$ as a subgraph. The *Turán number* $\mathrm{ex}(n, \mathcal{F})$ of $\mathcal{F}$ is the maximum number of edges in an $\mathcal{F}$-free r-graph on n vertices. The *Turán density* $\pi(\mathcal{F})$ of $\mathcal{F}$ is defined as $\pi(\mathcal{F}) = \lim_{n\to\infty} \mathrm{ex}(n, \mathcal{F})/\binom{n}{r}$. Define

$$\Pi_{\mathrm{fin}}^{(r)} := \{\pi(\mathcal{F}) \colon \mathcal{F} \text{ is a finite family of } r\text{-graphs}\}, \quad \text{and}$$
$$\Pi_{\infty}^{(r)} := \{\pi(\mathcal{F}) \colon \mathcal{F} \text{ is a (possibly infinite) family of } r\text{-graphs}\}.$$

[*]Email: jfhou@fzu.edu.cn
[†]Email: hucaiyun.fzu@gmail.com
[‡]Email: wu@hunnu.edu.cn

The study of $\mathrm{ex}(n, \mathcal{F})$ is perhaps the central topic in extremal combinatorics. Much is known about $\mathrm{ex}(n, \mathcal{F})$ when $r = 2$, and one of the most famous results in this regard is Turán's theorem, which states that for every integer $\ell \geq 2$ the Turán number $\mathrm{ex}(n, K_{\ell+1})$ is uniquely achieved by the balanced ℓ-partite graph on n vertices. The classical Erdős–Stone–Simonovits theorem [7, 9] shows that

$$\Pi_{\mathrm{fin}}^{(2)} = \Pi_{\infty}^{(2)} = \left\{ \frac{m-1}{m} : m = 1, 2, \ldots \right\}.$$

Unfortunately, little is known about $\Pi_{\mathrm{fin}}^{(r)}$ and $\Pi_{\infty}^{(r)}$ for $r \geq 3$. This is perhaps a barrier in determining Turán densities of hypergrahs. Erdős [6] proved that $\Pi_{\infty}^{(r)} \cap (0, r!/r) = \emptyset$. Chung and Graham [5] conjectured that $\Pi_{\mathrm{fin}}^{(r)}$ consists of rational numbers only. This conjecture was disproved by Baber and Talbot [1], who demonstrated that even a family of just three forbidden 3-graphs can have an irrational Turán density. In 2014, Pikhurko [17] showed that $\Pi_{\infty}^{(r)}$ has cardinality of the continuum for every $r \geq 3$. In [1], Baber and Talbot posed the following open problem.

lbot-irrational

Problem 1.1. *Does there exist single r-uniform hypergraphs with irrational Turán densities?*

Problem 1.1 was first addressed by Yan and Peng [21], who provided an explicit example of a single 3-graph with irrational Turán density. Shortly afterwards, Wu [20] presented another such example of a single forbidden 4-graph with the same property. More recently, Kamčev, Letzter and Pokrovskiy [15] determined the Turán density of 3-uniform tight ℓ-cycle $\mathcal{C}_\ell$ for sufficiently large ℓ not divisible by 3. This was subsequently extended by Bondnár, León, Liu and Pikhurko [3] to all $\ell \geq 7$ not divisible by 3, establishing that $\pi(\mathcal{C}_\ell) = 2\sqrt{3} - 3$. The main contribution of this paper is to establish the existence of a single 5-graph with irrational Turán density.

Before stating our main result, we introduce some necessary notation. Let $M_t^{(r)}$ be the r-graph formed by t disjoint edges. Let $\mathcal{G}$ be a r-graph. We say that $\{u, v\} \subseteq V(\mathcal{G})$ is *covered* in $\mathcal{G}$ if there exists $e \in \mathcal{G}$ containing u and v. We say $\mathcal{G}$ is *2-covered* if every pair of vertices is covered. The *extension* of $\mathcal{G}$, denoted by $\mathcal{H}^{\mathcal{G}}$, is obtained from $\mathcal{G}$ by adding C_{ij} of $r-2$ new vertices for each uncovered pairs $\{v_i, v_j\}$ in $\mathcal{G}$ and adding the edge $\{v_i, v_j\} \cup C_{ij}$.

y-M_{3,4}^{(5)}

Theorem 1.2. *For $k \in \{3, 4\}$, let $\mathcal{H}_k$ be the extension of $M_k^{(5)}$. Then*

$$\pi(\mathcal{H}_3) = \frac{9}{2048}, \quad \textit{and} \quad \pi(\mathcal{H}_4) = \frac{204 - 14\sqrt{21}}{28125}.$$

1.2 Lagrangian densities

We prove Theorem 1.2 using the classical Lagrangian method, posed by Motzkin and Straus [16].

Definition 1.3. *Let $\mathcal{G}$ be an r-graph on vertex set $[n]$. Define the* Lagrangian *function of G as*

$$\lambda(\mathcal{G}, \boldsymbol{x}) = \sum_{e \in \mathcal{G}} \prod_{i \in e} x_i,$$

where $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in [0, \infty)^n$. The Lagrangian *of $\mathcal{G}$ is defined as*

$$\lambda(\mathcal{G}) = \max\{\lambda(\mathcal{G}, \boldsymbol{x}) : \boldsymbol{x} \in \Delta_n\},$$

where $\Delta_n = \{\boldsymbol{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1, x_i \geq 0$ for each $i \in [n]\}$.

We call $\boldsymbol{x} \in \Delta_n$ *optimal* if $\lambda(\mathcal{G}) = \lambda(\mathcal{G}, \boldsymbol{x})$.

Given a family $\mathcal{F}$ of r-graphs, the *Lagrangian density* of $\mathcal{F}$ is defined to be

$$\pi_\lambda(\mathcal{F}) = r! \sup\{\lambda(\mathcal{G}) : \mathcal{G} \text{ is } \mathcal{F}\text{-free}\}.$$

The hypergraph Lagrangian was developed independently by Sidorenko [18] and by Frankl and Füredi [10], generalizing the earlier work of Motzkin and Straus [16] and Zykov [22]. The Lagrangian density problem has been studied in the recent years, strongly connected to the Turán problems.

extension **Proposition 1.4** ([4, 18, 19])**.** *For an r-graph F, the following hold:*

(i) $\pi(F) \le \pi_\lambda(F)$*;*

(ii) $\pi(H^F) = \pi_\lambda(F)$*. In particular,* $\pi(F) = \pi_\lambda(F)$ *if F is 2-covered.*

Most results on hypergraph Lagrangian densities are focused on "tree"-like hypergraphs. For instance, for $r \ge 3$, let H^r be the r-graph on $r+1$ vertices consisting of two edges that share exactly $r-1$ vertices. The Lagrangian density $\pi_\lambda(H^r)$ was determined by Sidorenko [18] for $r \in 3, 4$, and by Frankl and Főredi [11] for $r \in 5, 6$. All remaining cases are still open. For sufficently larger t, let T be a tree satisfying Erdős-Sós Conjecture, and let F be an r-graph obtained by joining $r-2$ fixed vertices into every edge of T. Sidorenko [19] proved that $\pi_\lambda(F) = \binom{t+r-3}{r}\frac{r!}{(t+r-3)^r}$. It is worth noting that the Lagrangian densities of these hypergraphs exhibit a profound connection to the Lagrangians of complete hypergraphs.

A natural problem is to determine the Lagrangian densities of matching $M_t^{(r)}$. The simplest case is when $t = 2$ (i.e., intersecting families). This was first investigated by Hefetz and Keevash [13], who proved that if $\mathcal{G} \ne K_5^{(3)}$ is a $M_2^{(3)}$-free 3-graph, then $\lambda(\mathcal{G}) \le \lambda(K_5^{(3)}) - 10^{-3}$. For $r \ge 4$, Bene Watts, Norin and Yepremyan [2] determined $\pi_\lambda(M_2^{(r)})$, confirming a conjecture of Hefetz and Keevash Hefetz and Keevash [13].

For general t. let $S_t^{(r)}(n)$ be the r-graph on vertex set $[n]$ with edge set $\{e \in \binom{[n]}{r} : e \cap [t] \ne \emptyset\}$. When the vertex set is infinite, we denote it by $S_t^{(r)}$. Clearly, both $S_{t-1}^{(r)}$ and $K_{rt-1}^{(r)}$ are $M_t^{(r)}$-free. Motivated by the classical Erdős Matching Conjecture [8], Wu [20] and Jiang, Peng and Wu [14] independently conjectured the following.

main-conjecture **Conjecture 1.5** (Jiang, Peng and Wu [14])**.**

$$\pi_\lambda(M_t^{(r)}) = r! \max\left\{\lambda(K_{rt-1}^{(r)}), \lambda(S_{t-1}^{(r)})\right\}.$$

Jiang, Peng and Wu [14] confirmed Conjecture 1.5 when $r = 3$. For $r \ge 4$, little is known. Recently, Wu [20] confirmed Conjecture 1.5 when $(r, t) = (4, 3)$. We determine $\pi_\lambda(M_t^{(5)})$ for $t \in \{3, 4\}$.

M_3^5 **Theorem 1.6.** *For $t \in \{3, 4\}$, $\pi_\lambda(M_t^{(5)}) = 5!\lambda(S_{t-1}^{(5)})$.*

Clearly, Theorem 1.2 follows from proposition 1.4 and Theorem 3.2.

This paper is organized as follows. In the following section, we give some useful lemmas which will be used in our main result. We investigate the Lagrangian densities of $M_3^{(5)}$ and $M_4^{(5)}$ in Sections 3 and 4, respectively.

2 Preliminaries

C:Preliminaries

In this section, we describe some notation and terminologies, and list some useful lemmas. For two positive integers n_1 and n_2 with $n_1 < n_2$, let $[n_1] := \{1, \ldots, n_1\}$ and $[n_1, n_2] := [n_2] \setminus [n_1 - 1]$. Let $\mathcal{G}$ be an r-graph and a subset $S \subseteq V(\mathcal{G})$. We use $\mathcal{G}[S]$ to denote the subgraph induced by S and let $\mathcal{G} \setminus S = \mathcal{G}[V(\mathcal{G}) \setminus S]$. We call $\mathcal{G}$ is *dense* if every proper subgraph $\mathcal{G}'$ of $\mathcal{G}$ satisfies $\lambda(\mathcal{G}') < \lambda(\mathcal{G})$. The following two facts are obvious.

subset **Fact 2.1.** *If $\mathcal{G}'$ is a subgraph of an r-graph $\mathcal{G}$, then $\lambda(\mathcal{G}') \le \lambda(\mathcal{G})$.*

covers pairs **Fact 2.2** (Frankl and Rödl [12])**.** *If an r-graph $\mathcal{G}$ is dense, then $\mathcal{G}$ is 2-covered.*

Let $\mathcal{G}$ be an r-graph with the vertex set $[n]$. For a vertex $i \in [n]$ the *link* of i is defined as

$$L_{\mathcal{G}}(i) = \{e \in \binom{[n] \setminus \{i\}}{r-1} : e \cup \{i\} \in \mathcal{G}\}.$$

Given $i, j \in [n]$, define

$$L_{\mathcal{G}}(j \setminus i) = \left\{ f \in \binom{[n] \setminus \{i,j\}}{r-1} : f \cup \{j\} \in \mathcal{G} \text{ and } f \cup \{i\} \notin \mathcal{G} \right\}.$$

The *compression* of j to i in $\mathcal{G}$ is defined as

$$\pi_{ij}(\mathcal{G}) = (\mathcal{G} \setminus \{f \cup \{j\} : f \in L_{\mathcal{G}}(j \setminus i)\}) \cup \{f \cup \{i\} : f \in L_{\mathcal{G}}(j \setminus i)\}.$$

We say that $\mathcal{G}$ is *left-compressed* if $\pi_{ij}(\mathcal{G}) = \mathcal{G}$ for all $1 \le i < j \le n$.

x_i=x_j **Lemma 2.3** (Frankl and Rödl [12])**.** *Let $\mathcal{G}$ be an r-graph on the vertex set $[n]$. If $L_{\mathcal{G}}(j \setminus i) = L_{\mathcal{G}}(i \setminus j)$, then there is an optimal weighting $\boldsymbol{x} \in \Delta_n$ such that $x_i = x_j$.*

A key ingredient of our proof is the following ingenious lemma established by Jiang, Peng, and Wu [14].

and compressed **Lemma 2.4** (Jiang, Peng and Wu [14])**.** *Let $\mathcal{G}$ be an $M_t^{(r)}$-free r-graph. Then there exists an $M_t^{(r)}$-free, dense and left-compressed r-graph G' with $|V(\mathcal{G}')| \le |V(\mathcal{G})|$ such that $\lambda(\mathcal{G}') \ge \lambda(\mathcal{G})$.*

Now we define some special left-compressed $M_2^{(5)}$-free 5-graph on the vertex set $[n]$. When there is no ambiguity, we will simply denote by $ijk\ell m$ instead of the edge $\{i, j, k, l, m\}$ of 5-graph. Let

- $\mathcal{H}_1 = \{1jk\ell m : 2 \le j < k << m \le n\}$.
- $\mathcal{H}_2 = \{1jk\ell m : 2 \le j \le 6, j < k < \ell < m \le n\} \cup \{23456\}$.
- $\mathcal{H}_3 = \{ijk\ell m : 1 \le i < j < k \le 7, k < \ell < m \le n\}$.
- $\mathcal{H}_4 = \{1jk\ell m : 2 \le j \le 5, j < k < \ell < m \le n\}$
  $\cup \{167\ell m : 8 \le l < m \le n\} \cup \{e \in \binom{[2,7]}{5} : 2 \in e\}$.
- $\mathcal{H}_5 = \{12k\ell m : 3 \le k < \ell < m \le n\} \cup \{i678m : 1 \le i \le 2, m \le n\}$
  $\cup \{ijk\ell m : 1 \le i \le 2, 3 \le j \le 5, j < k \le 7, k < \ell < m \le n\}$.

- $\mathcal{H}_6 = \{1jk\ell m : 2 \le j \le 5, j < k < \ell < m \le n\} \cup \{1678m : 9 \le m \le n\}$
  $\cup \{23k\ell m : 4 \le k < \ell < m \le n\}$.

d-M25-sturcture **Lemma 2.5.** *Let $\mathcal{G}$ be a left-compressed $M_2^{(5)}$-free 5-graph on the vertex set $[n]$ with $n \ge 10$. Then $\mathcal{G}$ is contained in $\mathcal{H}_i$ for some $i \in [6]$.*

Proof. We may assume that $\mathcal{G}[[2,n]] \neq \emptyset$ since otherwise $\mathcal{G} \subseteq \mathcal{H}_1$. It follows $\mathcal{G}$ is left-compressed that $23456 \in \mathcal{G}$. Since $\mathcal{G}$ is M_2^5-free, we have $178910 \notin \mathcal{G}$. Moreover, if $168910 \in \mathcal{G}$, then $23457 \notin \mathcal{G}$, and so $\mathcal{G} \subseteq \mathcal{H}_2$. In the following, assume that $168910 \notin \mathcal{G}$. Suppose that $ijk\ell m \in \mathcal{G}$ with $1 \le i < j < k < \ell < m \le n$. Then $j \le 6$. If $34567 \in \mathcal{G}$, then $128910 \notin \mathcal{G}$. This means $k \le 7$ and then $\mathcal{G} \subseteq \mathcal{H}_3$.

Suppose that $34567 \notin \mathcal{G}$. By the fact that $\mathcal{G}$ is left-compressed, we can get $1 \le i \le 2$. If $167910 \in \mathcal{G}$, then $23458 \notin \mathcal{G}$ and so $\mathcal{G} \in \mathcal{H}_4$. Otherwise, $167910 \notin \mathcal{G}$. In this case, if $24567 \in \mathcal{G}$, then $\{1,3,8,9,10\} \notin \mathcal{G}$ and so $\mathcal{G} \subseteq \mathcal{H}_5$. Otherwise, we have $\mathcal{G} \subseteq \mathcal{H}_6$. ∎

Using the same arguments of Lemma 15 in [20], we can get

G[[2,n]] **Lemma 2.6.** *Let $t \ge 3$ and $n \ge 5t$ be two integers. Let $\mathcal{G}$ be a left-compressed and $M_t^{(5)}$-free 5-graph on the vertex set $[n]$. If $\mathcal{G}[[2,n]]$ contains a copy of $M_{t-1}^{(5)}$, then $\mathcal{G}[[2,5t-4]]$ contains a copy of $M_{t-1}^{(5)}$.*

3 Lagrangian densities of $M_3^{(5)}$

n-densities-M-3

In this section, we determine the Lagrangian density of $M_3^{(5)}$.

Lagrangian-S25 **Lemma 3.1.** $\lambda(S_2^{(5)}) = \frac{9}{2048} (\approx 4.39 \times 10^{-3})$.

Proof. Let $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\}$ be an optimal weighting of $S_2^{(5)}(n)$. By Lemma 2.3, we may assume that

$$x_1 = x_2,\ x_3 = x_4 = \cdots = x_n.$$

Let $x_1 = x_2 = \frac{a}{2}$. Then $x_3 = x_4 = \cdots = x_n = \frac{1-a}{n-2}$. It is obvious that

$$\begin{aligned}\lambda\left(S_2^{(5)}(n), \boldsymbol{x}\right) &= \left(\frac{a}{2}\right)^2 \binom{n-2}{3} \left(\frac{1-a}{n-2}\right)^3 + \frac{a}{2}\binom{n-2}{4}\left(\frac{1-a}{n-2}\right)^4 \\ &= f(a) - \Theta\left(\frac{1}{n}\right),\end{aligned}$$

where $f(a) = \left(\frac{a}{2}\right)^2 \left(\frac{1-a}{6}\right)^3 + \frac{(1-a)^4}{24} a$. Hence, $\lambda\left(S_2^{(5)}(n)\right) = \max\limits_{0 \le a \le 1} \{f(a) - \Theta(1/n)\}$, and then

$$\lambda(S_2^{(5)}) = \lim_{n\to\infty} \lambda\left(S_2^{(5)}(n)\right) = \max_{0 \le a \le 1} \{f(a)\} = f\left(\frac{1}{4}\right) = \frac{9}{2048},$$

completing the proof. ∎

M_3^5 **Theorem 3.2.** *We have $\pi_\lambda(M_3^{(5)}) = 5!\lambda(S_2^{(5)})$.*

Proof. It suffices to show $\pi_\lambda(M_3^{(5)}) \le 5!\lambda(S_2^{(5)})$. Let $\mathcal{G}$ be an $M_3^{(5)}$-free 5-graph on the vertex set $[n]$. By Lemma 2.4, we may assume that $\mathcal{G}$ is dense and left-compressed. If $n \le 14$, then by Fact 2.1, we have $\lambda(\mathcal{G}) \le \lambda(K_{14}^{(5)}) = \binom{14}{5}\frac{1}{14^5} < \lambda(S_2^{(5)})$. In the following, suppose that $n \ge 15$. We divide our proof into the following two cases.

Case 1: $\mathcal{G} \setminus \{1\}$ is $M_2^{(5)}$-free.

Let $\mathcal{G}_1 = \{1jk\ell m : 2 \le j < k < \ell < m \le n\}$. By Lemma 2.5, $\mathcal{G} \setminus \{1\}$ is contained in a copy of $\mathcal{H}_i$ for some $i \in [6]$. If $i = 1$, then $\mathcal{G}$ is contained in a copy of $S_2^{(5)}(n)$ and $\lambda(\mathcal{G}) \le \lambda(S_2^{(5)}(n))$ by Fact 2.1. In the following, suppose that $i \ge 2$.

Subcase 1.1. $\mathcal{G} \setminus \{1\}$ is contained in a copy of $\mathcal{H}_2$.

This means $\mathcal{G}$ is in a copy of

$$\mathcal{G}_{12} = \mathcal{G}_1 \cup \{2jk\ell m : 3 \le j \le 7, j < k < \ell < m \le n\} \cup \{34567\}.$$

Let $x_1 = a$, $x_2 = b$, $\sum_{i=3}^{7} x_i = c$ and $\sum_{i=8}^{n} x_i = d$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. By Lemma 2.3, we may assume that $x_3 = \cdots = x_7 = \frac{c}{5}$ and $x_8 = \cdots = x_n = \frac{d}{n-7}$. Taking edges of the form $\{2jk\ell m : 3 \le j \le 7, j < k < \ell < m \le n\}$ as an example, the contribution of these edges to $\lambda(\mathcal{G}_{12}, \boldsymbol{x})$ does not exceed

$$\begin{aligned}
&b\left(\binom{5}{4}\left(\frac{c}{5}\right)^4 + \binom{5}{3}\left(\frac{c}{5}\right)^3\binom{n-7}{1}\frac{d}{n-7} + \binom{5}{2}\left(\frac{c}{5}\right)^2\binom{n-7}{2}\left(\frac{d}{n-7}\right)^2 + c\binom{n-7}{3}\left(\frac{d}{n-7}\right)^3\right)\\
&\le b\left(\binom{5}{4}\left(\frac{c}{5}\right)^4 + \binom{5}{3}\left(\frac{c}{5}\right)^3 d + \binom{5}{2}\left(\frac{c}{5}\right)^2\frac{d^2}{2} + c\frac{d^3}{6}\right).
\end{aligned}$$

Let

$$X = \binom{5}{4}\left(\frac{c}{5}\right)^4 + \binom{5}{3}\left(\frac{c}{5}\right)^3 d + \binom{5}{2}\left(\frac{c}{5}\right)^2\frac{d^2}{2} + c\frac{d^3}{6},$$

and

$$f_{12} = a\left[b\left(\binom{5}{3}\left(\frac{c}{5}\right)^3 + \binom{5}{2}\left(\frac{c}{5}\right)^2 d + c\frac{d^2}{2} + \frac{d^3}{6}\right) + X + \frac{d^4}{24}\right] + bX + \left(\frac{c}{5}\right)^5.$$

Then

$$\lambda(\mathcal{G}_{12}) \le \max\{f_{12} : a + b + c + d = 1 \text{ and } a, b, c, d \ge 0\}.$$

Using LINGO, we obtain that $\lambda(\mathcal{G}) \le \lambda(\mathcal{G}_{12}) \approx 3.66902 \times 10^{-3} < \lambda(S_2^{(5)})$.

Subcase 1.2. $\mathcal{G} \setminus \{1\}$ is contained in a copy of $\mathcal{H}_3$.

This means $\mathcal{G}$ is in a copy of

$$\mathcal{G}_{13} = \mathcal{G}_1 \cup \{ijk\ell m : 2 \le i < j < k \le 8, k < \ell < m \le n\}.$$

Let

$$\begin{aligned}
f_{13} =& a\left(\binom{7}{4}\left(\frac{b}{7}\right)^4 + \binom{7}{3}\left(\frac{b}{7}\right)^3 c + \binom{7}{2}\left(\frac{b}{7}\right)^2\frac{c^2}{2} + b\frac{c^3}{6} + \frac{c^4}{24}\right)\\
&+ \binom{7}{5}\left(\frac{b}{7}\right)^5 + \binom{7}{4}\left(\frac{b}{7}\right)^4 c + \binom{7}{3}\left(\frac{b}{7}\right)^3\frac{c^2}{2}.
\end{aligned}$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{13}) \le \max\{f_{13} : a+b+c=1 \text{ and } a,b,c \ge 0\}$$

by letting $x_1 = a$, $\sum_{i=2}^{8} x_i = b$ and $\sum_{i=9}^{n} x_i = c$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \le \lambda(\mathcal{G}_{13}) \approx 3.59 \times 10^{-3} < \lambda(S_2^{(5)})$.

Subcase 1.3. $\mathcal{G} \setminus \{1\}$ is contained in a copy of $\mathcal{H}_4$.

In this subcase,

$$\begin{aligned}\mathcal{G} \subseteq \mathcal{G}_{14} =& \mathcal{G}_1 \cup \{2jk\ell m : 3 \le j \le 6, j < k < \ell < m \le n\} \\ & \cup \{278\ell m : 9 \le \ell < m \le n\} \cup \{e \in \binom{[3,8]}{5} : 3 \in e\}.\end{aligned}$$

Let

$$\begin{aligned}f_{14} =& a\left\{b\left[c\left(\binom{3}{2}\left(\frac{d}{3}\right)^2 + de + df + \left(\frac{e}{2}\right)^2 + ef + \frac{f^2}{2}\right) + X\right] + cX + Y + \left(\frac{e}{2}\right)^2\frac{f^2}{2} + e\frac{f^3}{6} + \frac{f^4}{24}\right\} \\ & + b(cX+Y) + b\left(\frac{e}{2}\right)^2\frac{f^2}{2} + c\left(\left(\frac{d}{3}\right)^3 e + \binom{3}{2}\left(\frac{d}{3}\right)^2\left(\frac{e}{2}\right)^2\right),\end{aligned}$$

where

$$X = \left(\frac{d}{3}\right)^3 + \binom{3}{2}\left(\frac{d}{3}\right)^2(e+f) + d\left(\left(\frac{e}{2}\right)^2 + ef + \frac{f^2}{2}\right) + \left(\frac{e}{2}\right)^2 f + e\frac{f^2}{2} + \frac{f^3}{6},$$

and

$$Y = \left(\frac{d}{3}\right)^3(e+f) + \binom{3}{2}\left(\frac{d}{3}\right)^2\left(\left(\frac{e}{2}\right)^2 + ef + \frac{f^2}{2}\right) + d\left(\left(\frac{e}{2}\right)^2 f + e\frac{f^2}{2} + \frac{f^3}{6}\right).$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{14}) = \max\{f_{14} : a+b+c+d+e+f=1 \text{ and } a,b,c,d,e,f \ge 0\}$$

by letting $x_1 = a$, $x_2 = b$, $x_3 = c$, $\sum_{i=4}^{6} x_i = d$, $x_7 + x_8 = e$ and $\sum_{i=9}^{n} x_i = f$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \le \lambda(\mathcal{G}_{14}) \approx 3.61 \times 10^{-3} < \lambda(S_2^{(5)})$.

Subcase 1.4: $\mathcal{G} \setminus \{1\}$ is contained in a copy of $\mathcal{H}_5$.

Again, we have

$$\begin{aligned}\mathcal{G} \subseteq \mathcal{G}_{15} =& \mathcal{G}_1 \cup \{23k\ell m : 4 \le k < \ell < m \le n\} \\ & \cup \{ijk\ell m : 2 \le i \le 3, 4 \le j \le 6, j < k \le 8, k < \ell < m \le n\} \\ & \cup \{i789m : 2 \le i \le 3, 10 \le m \le n\}.\end{aligned}$$

Let

$$\begin{aligned}f_{15} =& a\left[\left(\frac{b}{2}\right)^2\left(\binom{3}{2}\left(\frac{c}{3}\right)^2 + c(d+e+f) + \left(\frac{d}{2}\right)^2 + de + df + ef + \frac{f^2}{2}\right) + bX + Y + ce\frac{f^2}{2}\right. \\ & \left. + c\frac{f^3}{6} + \left(\frac{d}{2}\right)^2\left(ef + \frac{f^2}{2}\right) + d\left(e\frac{f^2}{2} + \frac{f^3}{6}\right) + e\frac{f^3}{6} + \frac{f^4}{24}\right] + \left(\frac{b}{2}\right)^2 X + bY + b\left(\frac{d}{2}\right)^2 ef,\end{aligned}$$

where

$$X = \left(\frac{c}{3}\right)^3 + \binom{3}{2}\left(\frac{c}{3}\right)^2 (d+e+f) + c\left(\left(\frac{d}{2}\right)^2 + de + df + ef + \frac{f^2}{2}\right) + \left(\frac{d}{2}\right)^2 (e+f) + d\left(ef + \frac{f^2}{2}\right) + e\frac{f^2}{2} + \frac{f^3}{6},$$

$$Y = \left(\frac{c}{3}\right)^3 (d+e+f) + \binom{3}{2}\left(\frac{c}{3}\right)^2 \left(\left(\frac{d}{2}\right)^2 + de + df + ef + \frac{f^2}{2}\right) + c\left(\left(\frac{d}{2}\right)^2 (e+f) + d(ef + \frac{f^2}{2})\right).$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{15}) = \max\{f_{15} : a+b+c+d+e+f = 1 \text{ and } a,b,c,d,e,f \geq 0\}$$

by letting $x_1 = a$, $x_2 + x_3 = b$, $\sum_{i=4}^{6} x_i = c$, $x_7 + x_8 = d$, $x_9 = e$ and $\sum_{i=10}^{n} x_i = f$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \leq \lambda(\mathcal{G}_{15}) \approx 3.44 \times 10^{-3} < \lambda(S_2^{(5)})$.

Subcase 1.5: $\mathcal{G} \setminus \{1\}$ is contained in a copy of $\mathcal{H}_6$.

We have

$$\begin{aligned}\mathcal{G} \subseteq \mathcal{G}_{16} =& \mathcal{G}_1 \cup \{2jk\ell m : 3 \leq j \leq 6, j < k < \ell < m \leq n\} \\ & \cup \{2789m : 10 \leq m \leq n\} \cup \{34k\ell m : 5 \leq k < \ell < m \leq n\}.\end{aligned}$$

Let

$$f_{16} = a\left[b\left(\left(\frac{c}{2}\right)^2 (d+e+f) + cX + Y\right) + Z + \left(\frac{e}{3}\right)^3 f + \binom{3}{2}\left(\frac{e}{3}\right)^2 \frac{f^2}{2} + e\frac{f^3}{6} + \frac{f^4}{24}\right] + bZ + b\left(\frac{e}{3}\right)^3 f,$$

where

$$X = \left(\frac{d}{2}\right)^2 + de + df + \binom{3}{2}\left(\frac{e}{3}\right)^2 + ef + \frac{f^2}{2},$$

$$Y = \left(\frac{e}{3}\right)^3 + \binom{3}{2}\left(\frac{e}{3}\right)^2 f + e\frac{f^2}{2} + \frac{f^3}{6},$$

and

$$Z = \left(\frac{c}{2}\right)^2 X + c\left(\left(\frac{d}{2}\right)^2 (e+f) + d\left(\binom{3}{2}\left(\frac{e}{3}\right)^2 + ef + \frac{f^2}{2}\right) + Y\right) + \left(\frac{d}{2}\right)^2 \left(\binom{3}{2}\left(\frac{e}{3}\right)^2 + ef + \frac{f^2}{2}\right) + dY.$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{16}) = \max\{f_{16} : a+b+c+d+e+f = 1 \text{ and } a,b,c,d,e,f \geq 0\}$$

by letting $x_1 = a$, $x_2 = b$, $x_3 + x_4 = c$, $x_5 + x_6 = d$, $\sum_{i=7}^{9} x_i = e$ and $\sum_{i=10}^{n} x_i = f$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \leq \lambda(\mathcal{G}_{16}) \approx 3.48 \times 10^{-3} < \lambda(S_2^{(5)})$.

Case 2: $M_2^{(5)} \subseteq \mathcal{G} \setminus \{1\}$.

By Lemma 2.6, we have $M_2^{(5)} \subseteq \mathcal{G}[[2,11]]$. Note that $\{1,12,13,14,15\} \notin \mathcal{G}$ as $\mathcal{G}$ is $M_3^{(5)}$-free. Thus for $j \le 11$, every element in $\{ijk\ell m : 1 \le i < j < k < \ell < m \le n\}$ is in $\mathcal{G}$.

Subcase 2.1: $\mathcal{G} \setminus \{1,2\}$ is $M_2^{(5)}$-free.

First, if $\mathcal{G} \setminus \{1,2\}$ is contained in a copy of $\mathcal{H}_1$ or $\mathcal{H}_2$, then

$$\mathcal{G} \subseteq \mathcal{G}_{211} = \{ijk\ell m : i \in [3], i < j \le 11, j < k < \ell < m \le n\} \cup \binom{[4,11]}{5}.$$

Let

$$\begin{aligned} f_{211} = &\left(\frac{a}{3}\right)^3 \left(\binom{8}{2}\left(\frac{b}{8}\right)^2 + bc + \frac{c^2}{2}\right) + \binom{3}{2}\left(\frac{a}{3}\right)^2 \left[\binom{8}{3}\left(\frac{b}{8}\right)^3 + \binom{8}{2}\left(\frac{b}{8}\right)^2 c + b\frac{c^2}{2} + \frac{c^3}{6}\right] \\ &+ a\left[\binom{8}{4}\left(\frac{b}{8}\right)^4 + \binom{8}{3}\left(\frac{b}{8}\right)^3 c + \binom{8}{2}\left(\frac{b}{8}\right)^2 \frac{c^2}{2} + b\frac{c^3}{6}\right] + \binom{8}{5}\left(\frac{b}{8}\right)^5. \end{aligned}$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{211}) = \max\{f_{211} : a+b+c = 1 \text{ and } a,b,c \ge 0\}$$

by letting $\sum_{i=1}^{3} x_i = a$, $\sum_{i=4}^{11} x_i = a$ and $\sum_{i=12}^{n} x_i = c$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \le \lambda(\mathcal{G}_{211}) \approx 4.15 \times 10^{-3} < \lambda(S_2^{(5)})$.

Then, if $\mathcal{G} \setminus \{1,2\}$ is contained in a copy of $\mathcal{H}_3$, then

$$\begin{aligned} \mathcal{G} \subseteq \mathcal{G}_{212} = &\{ijk\ell m : i \in [2], i < j \le 11, j < k < \ell < m \le n\} \\ &\cup \{ijk\ell m : 3 \le i < j < k \le 9, k < \ell < m \le n\}. \end{aligned}$$

Let

$$\begin{aligned} f_{212} = &\left(\frac{a}{2}\right)^2 \left[\binom{7}{3}\left(\frac{b}{7}\right)^3 + \binom{7}{2}\left(\frac{b}{7}\right)^2 (c+d) + bX + Y\right] + a\left[\binom{7}{4}\left(\frac{b}{7}\right)^4 \right. \\ &\left. + \binom{7}{3}\left(\frac{b}{7}\right)^3 (c+d) + \binom{7}{2}\left(\frac{b}{7}\right)^2 X + bY + \left(\frac{c}{2}\right)^2 \frac{d^2}{2} + c\frac{d^3}{6}\right] \\ &+ \binom{7}{5}\left(\frac{b}{7}\right)^5 + \binom{7}{4}\left(\frac{b}{7}\right)^4 (c+d) + \binom{7}{3}\left(\frac{b}{7}\right)^3 X. \end{aligned}$$

where

$$X = \left(\frac{c}{2}\right)^2 + cd + \frac{d^2}{2} \text{ and } Y = \left(\frac{c}{2}\right)^2 d + c\frac{d^2}{2} + \frac{d^3}{6}.$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{212}) = \max\{f_{212} : a+b+c+d = 1 \text{ and } a,b,c,d \ge 0\}$$

by letting $x_1 + x_2 = a$, $\sum_{i=3}^{9} x_i = b$, $x_{10} + x_{11} = c$ and $\sum_{i=12}^{n} x_i = d$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \le \lambda(\mathcal{G}_{212}) \approx 4.05 \times 10^{-3} < \lambda(S_2^{(5)})$.

At last, if $\mathcal{G} \setminus \{1,2\}$ is contained in a copy of $\mathcal{H}_i$ for some $i \in \{4,5,6\}$, then

$$\begin{aligned} \mathcal{G} \subseteq \mathcal{G}_{213} = &\{ijk\ell m : i \in [2], i < j \le 11, j < k < \ell < m \le n\} \\ &\cup \{ijk\ell m : i \in [3,4], i < j \le 8, j < k < \ell < m \le n\}. \end{aligned}$$

Let

$$f_{213} = \left(\frac{a}{2}\right)^2 \left(\left(\frac{b}{2}\right)^2 (c+d+e) + bX + Y \right) + a \Bigg[\left(\frac{b}{2}\right)^2 X + bY + Z + \left(\frac{d}{3}\right)^3 e + \binom{3}{2}\left(\frac{d}{3}\right)^2 \frac{e^2}{2} + d\frac{e^3}{6} \Bigg] + \left(\frac{b}{2}\right)^2 Y + bZ.$$

where

$$X = \binom{4}{2}\left(\frac{c}{4}\right)^2 + c(d+e) + \binom{3}{2}\left(\frac{d}{3}\right)^2 + de + \frac{e^2}{2},$$

$$Y = \binom{4}{3}\left(\frac{c}{4}\right)^3 + \binom{4}{2}\left(\frac{c}{4}\right)^2 (d+e) + c\left(\binom{3}{2}\left(\frac{d}{3}\right)^2 + de + \frac{e^2}{2} \right) + \left(\frac{d}{3}\right)^3 + \binom{3}{2}\left(\frac{d}{3}\right)^2 e + d\frac{e^2}{2} + \frac{e^3}{6}$$

and

$$Z = \left(\frac{c}{4}\right)^4 + \binom{4}{3}\left(\frac{c}{4}\right)^3 (d+e) + \binom{4}{2}\left(\frac{c}{4}\right)^2 \left(\binom{3}{2}\left(\frac{d}{3}\right)^2 + de + \frac{e^2}{2} \right) + c\left(\left(\frac{d}{3}\right)^3 + \binom{3}{2}\left(\frac{d}{3}\right)^2 e + d\frac{e^2}{2} + \frac{e^3}{6} \right).$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{213}) = \max\{f_{213} : a+b+c+d+e = 1 \text{ and } a,b,c,d,e \geq 0\}$$

by letting $x_1 + x_2 = a$, $x_3 + x_4 = b$, $\sum_{i=5}^{8} x_i = c$, $\sum_{i=9}^{11} x_i = d$ and $\sum_{i=12}^{n} x_i = e$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \leq \lambda(\mathcal{G}_{213}) \approx 4.31 \times 10^{-3} < \lambda(S_2^{(5)})$.

Subcase 2.2: $M_2^{(5)} \subseteq \mathcal{G} \setminus \{1,2\}$.

By Lemma 2.6, we have $M_2^{(5)} \subseteq \mathcal{G}[[3,12]]$. Clearly, $\{1,2,13,14,15\} \notin \mathcal{G}$. Thus for $k \leq 12$ every element in $\{ijk\ell m : 1 \leq i < j < k < \ell < m \leq n\}$ is in $\mathcal{G}$. Due to $\mathcal{G}$ is dense, we have $\{1,2,3,n-1,n\} \in \mathcal{G}$ by Fact 2.1, and so $\mathcal{G}[[4,n]]$ is $M_2^{(5)}$-free. Therefore $\lambda(\mathcal{G}[[4,n]]) \leq \lambda(S_1^{(5)})$. This means $\mathcal{G}$ is in a copy of

$$\mathcal{G}_{22} = \{ijk\ell m : i \in [3], i < j \leq 11, j < k \leq 12, k < \ell < m \leq n\} \cup \mathcal{G}[[4,n]].$$

Let

$$f_{22} = \left(\frac{a}{3}\right)^3 \left(\binom{8}{2}\left(\frac{b}{8}\right)^2 + bc + bd + cd + \frac{d^2}{2} \right) + \binom{3}{2}\left(\frac{a}{3}\right)^2 \Bigg[\binom{8}{3}\left(\frac{b}{8}\right)^3 + \binom{8}{2}\left(\frac{b}{8}\right)^2 (c+d) + b\left(cd + \frac{d^2}{2}\right) + c\frac{d^2}{2} \Bigg] + a\Bigg[\binom{8}{4}\left(\frac{b}{8}\right)^4 + \binom{8}{3}\left(\frac{b}{8}\right)^3 (c+d) + \binom{8}{2}\left(\frac{b}{8}\right)^2 \left(cd + \frac{d^2}{2}\right) + bc\frac{d^2}{2} \Bigg] + \lambda(S_1^{(5)})(b+c+d)^5.$$

Then by Lemma 2.3,

$$\lambda(\mathcal{G}_{22}) = \max\{f_{22} : a+b+c+d = 1 \text{ and } a,b,c,d \geq 0\}$$

by letting $\sum_{i=1}^{3} x_i = a$, $\sum_{i=4}^{11} x_i = b$, $x_{12} = c$ and $\sum_{i=13}^{n} x_i = d$ for $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\} \in \Delta_n$. Using LINGO, we obtain that $\lambda(\mathcal{G}) \leq \lambda(\mathcal{G}_{22}) \approx 4.29 \times 10^{-3} < \lambda(S_2^{(5)})$. ∎

4 Lagrangian densities of $M_4^{(5)}$

n-densities-M-4

In this section, we determine the Lagrangian density of $M_4^{(5)}$. Through the process of the proof of Theorem 3.2, we can obtain the following lemma.

0.00366902 **Lemma 4.1.** *Let $\mathcal{G}$ be a left-compressed $M_2^{(5)}$-free 5-graph on vertex set $[n]$ with $n \geq 10$. If $\mathcal{G} \setminus \{1\}$ is $M_2^{(5)}$-free and $M_1^{(5)} \subseteq \mathcal{G} \setminus \{1,2\}$, then $\lambda(\mathcal{G}) \leq 3.66902 \times 10^{-3}$.*

Lemma 4.2. $\lambda(S_3^{(5)}) = \frac{204-14\sqrt{21}}{28125} (\approx 4.97223 \times 10^{-3})$.

Proof. Let $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\}$ be an optimal weighting of $S_2^{(4)}(n)$. By Lemma 2.3, we may assume that

$$x_1 = x_2 = x_3,\ x_4 = x_5 = \cdots = x_n.$$

Let $x_1 = x_2 = x_3 = \frac{a}{3}$. Then $x_4 = x_5 = \cdots = x_n = \frac{1-a}{n-3}$. It is obvious that

$$\begin{aligned}\lambda\left(S_3^{(5)}(n), \boldsymbol{x}\right) &= \left(\frac{a}{3}\right)^3 \binom{n-3}{2} \left(\frac{1-a}{n-3}\right)^2 + 3 \cdot \left(\frac{a}{3}\right)^2 \binom{n-3}{3} \left(\frac{1-a}{n-3}\right)^3 + 3 \cdot \frac{a}{3} \binom{n-3}{4} \left(\frac{1-a}{n-3}\right)^4 \\ &= f(a) - \Theta\left(\frac{1}{n}\right),\end{aligned}$$

where $f(a) = \left(\frac{a}{3}\right)^3 \frac{(1-a)^2}{2} + 3\left(\frac{a}{3}\right)^2 \frac{(1-a)^3}{6} + a\frac{(1-a)^4}{24}$. Hence, $\lambda\left(S_3^{(5)}(n)\right) = \max\limits_{0 \leq a \leq 1} \{f(a) - \Theta(1/n)\}$, and then

$$\lambda(S_3^{(5)}) = \lim_{n\to\infty} \lambda\left(S_2^{(5)}(n)\right) = \max_{0 \leq a \leq 1} \{f(a)\}.$$

Note that

$$f'(a) = \frac{(1-a)^4}{24} - \frac{a(1-a)^3}{18} - \frac{a^2(1-a)^2}{9} - \frac{a^3(1-a)}{27}.$$

The equation $f'(a) = 0$ yields two stationary points: $a = \frac{6-\sqrt{21}}{5}$ or $a = 1$. It is straightforward to verify that

$$f(a) \leq f\left(\frac{6-\sqrt{21}}{5}\right) = \frac{204 - 14\sqrt{21}}{3^2 \cdot 5^5},$$

completing the proof. ∎

M_4^5 **Theorem 4.3.** $\pi_\lambda(M_4^{(5)}) = 5!\lambda(S_3^{(5)})$.

Proof. It suffices to show $\pi_\lambda(M_4^{(5)}) \leq 5!\lambda(S_3^{(5)})$. The proof idea is similar to that of Theorem 3.2. Let $\mathcal{G}$ be an $M_4^{(5)}$-free 5-graph on the vertex set $[n]$. By Lemma 2.4, we may assume that $\mathcal{G}$ is dense and left-compressed. If $n \leq 19$, by Fact 2.1, we have $\lambda(\mathcal{G}) \leq \lambda(K_{19}^{(5)}) = \binom{19}{5}\frac{1}{19^5} < \lambda(S_3^{(5)})$. In the following, suppose that $n \geq 20$. For simplicity, if $ijk\ell m$ is an edge of $\mathcal{G}$, then we always suppose that $i < j < k < \ell < m$.

Assume that $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\}$ is an optimal weighting of $\mathcal{G}$ chosen via Lemma 2.3. In the following, we set a, b, c, d, e, f, g be variables exhibited in Table 1. Each of these consists the sum of some x_i. In Table 1, f_x is a function with respect to some (or all) variables in $\{a, b, c, d, e, f, g\}$, and the last column of Table 1 lists the maximum of f_x using LINGO. Clearly, all values are less than $\lambda(S_3^{(5)})$. The main task in the following is

Case	a	b	c	d	e	f	g	f_x	max of $f_x \times 10^3$
1.2.1	x_1	$\sum_{i=2}^{4} x_i$	$\sum_{i=5}^{12} x_i$	$\sum_{i=13}^{n} x_i$	–	–	–	$f_{121(a)}$	4.66
	x_1	x_2	$\sum_{i=3}^{12} x_i$	$\sum_{i=13}^{n} x_i$	–	–	–	$f_{121(b)}$	4.9720
1.2.2	x_1	$\sum_{i=2}^{13} x_i$	$\sum_{i=14}^{n} x_i$	–	–	–	–	f_{122}	4.65
2.1.1	$\sum_{i=1}^{4} x_i$	$\sum_{i=5}^{17} x_i$	$\sum_{i=18}^{n} x_i$	–	–	–	–	f_{211}	4.86
2.1.2	$\sum_{i=1}^{3} x_i$	$x_4 + x_5$	$\sum_{i=6}^{14} x_i$	$\sum_{i=15}^{17} x_i$	$\sum_{i=18}^{n} x_i$	–	–	$f_{212(a)}$	4.968
	$\sum_{i=1}^{3} x_i$	$\sum_{i=4}^{15} x_i$	$x_{16} + x_{17}$	$\sum_{i=18}^{n} x_i$	–	–	–	$f_{212(b)}$	4.94
2.2.1	$\sum_{i=1}^{4} x_i$	$\sum_{i=5}^{16} x_i$	$\sum_{i=17}^{n} x_i$	–	–	–	–	$f_{221(a)}$	4.73
	$x_1 + x_2$	$\sum_{i=3}^{16} x_i$	$\sum_{i=17}^{n} x_i$	–	–	–	–	$f_{221(b)}$	4.91
2.2.2.1	$x_1 + x_2$	$x_3 + x_4$	$\sum_{i=5}^{11} x_i$	$x_{12} + x_{13}$	$\sum_{i=14}^{16} x_i$	$\sum_{i=17}^{n} x_i$	–	$f_{2221(a)}$	4.82
	$x_1 + x_2$	$\sum_{i=3}^{5} x_i$	x_6	$\sum_{i=7}^{13} x_i$	$\sum_{i=14}^{16} x_i$	$\sum_{i=17}^{n} x_i$	–	$f_{2221(b_1)}$	4.91
	$x_1 + x_2$	$x_3 + x_4$	$x_5 + x_6$	$\sum_{i=7}^{9} x_i$	$\sum_{i=10}^{13}$	$\sum_{i=14}^{16} x_i$	$\sum_{i=17}^{n} x_i$	$f_{2221(b_2)}$	4.74
2.2.2.2	$x_1 + x_2$	$\sum_{i=3}^{14} x_i$	$x_{15} + x_{16}$	$\sum_{i=17}^{n} x_i$	–	–	–	f_{2222}	4.90

Table 1: Variables and functions table

to define a proper f_x in different cases whose maximum is the Lagrangian of $\mathcal{G}$. We divide our proof into the following two cases.

Case 1: $\mathcal{G} \setminus \{1\}$ is $M_3^{(5)}$-free.

Subcase 1.1: $\mathcal{G} \setminus \{1,2\}$ is $M_2^{(5)}$-free.

If $\mathcal{G} \setminus \{1,2,3\}$ is $M_1^{(5)}$-free, we have $\mathcal{G} \subseteq S_3^{(5)}$ and we are done. Now suppose that $M_1^{(5)} \subseteq \mathcal{G} \setminus \{1,2,3\}$. By Lemma 4.1 $\lambda(\mathcal{G} \setminus \{1\}) \le 3.66902 \times 10^{-3}$. Let $x_1 = a$ and $\sum_{i=2}^{n} x_i = 1 - a$. Using LINGO, we obtain

$$\lambda(\mathcal{G}) \le \max_{0 \le a \le 1} \left\{ a\frac{(1-a)^4}{24} + \lambda(\mathcal{G} \setminus \{1\})(1-a)^5 \right\} \le 4.94 \times 10^{-3} < \lambda(S_3^{(5)}).$$

Subcase 1.2: $M_2^{(5)} \subseteq \mathcal{G} \setminus \{1,2\}$.

By Lemma 2.6, we have $M_2^{(5)} \subseteq \mathcal{G}[[3,12]]$. It follows from $\mathcal{G} \setminus \{1\}$ is $M_3^{(5)}$-free that $\{2,13,14,15,16\} \notin \mathcal{G}$. This means

$$\text{if } i \ge 2, \quad \text{then } j \le 12 \text{ for } ijk\ell m \in \mathcal{G}. \tag{1}$$ 1.2

Subcase 1.2.1: $\mathcal{G} \setminus \{1,2,3\}$ is $M_2^{(5)}$-free.

If $\mathcal{G} \setminus \{1,2,3,4\}$ is $M_1^{(5)}$-free, then by (1), we have

$$\lambda(\mathcal{G}) \le f_{121(a)},$$

where

$$f_{121(a)} = a\left(\left(\frac{b}{3}\right)^3 (c+d) + \binom{3}{2}\left(\frac{b}{3}\right)^2 X + bY + Z\right) + \left(\frac{b}{3}\right)^3 X + \binom{3}{2}\left(\frac{b}{3}\right)^2 Y + bZ$$

with

$$X = \binom{8}{2}\left(\frac{c}{8}\right)^2 + cd + \frac{d^2}{2}, \quad Y = \binom{8}{3}\left(\frac{c}{8}\right)^3 + \binom{8}{2}\left(\frac{c}{8}\right)^2 d + c\frac{d^2}{2} + \frac{d^3}{6},$$

and

$$Z=\binom{8}{4}\left(\frac{c}{8}\right)^4+\binom{8}{3}\left(\frac{c}{8}\right)^3 d+\binom{8}{2}\left(\frac{c}{8}\right)^2\frac{d^2}{2}+c\frac{d^3}{6}.$$

Otherwise, $M_1^{(5)}\subseteq\mathcal{G}\setminus\{1,2,3,4\}$. By Lemma 4.1, we have $\lambda(\mathcal{G}\setminus\{1,2\})\leq 3.66902\times 10^{-3}$, which together with (1) yields

$$\lambda(\mathcal{G})\leq f_{121(b)},$$

where

$$\begin{aligned}f_{121(b)}=&a\left[b\left(\binom{10}{3}\left(\frac{c}{10}\right)^3+\binom{10}{2}\left(\frac{c}{10}\right)^2 d+c\frac{d^2}{2}+\frac{d^3}{6}\right)+X+\frac{d^4}{24}\right]\\&+bX+\lambda(\mathcal{G}\setminus\{1,2\})(c+d)^5\end{aligned}$$

with

$$X=\binom{10}{4}\left(\frac{c}{10}\right)^4+\binom{10}{3}\left(\frac{c}{10}\right)^3 d+\binom{10}{2}\left(\frac{c}{10}\right)^2\frac{d^2}{2}+c\frac{d^3}{6}.$$

Subcase 1.2.2: $M_2^{(5)}\subseteq\mathcal{G}\setminus\{1,2,3\}$.

By Lemma 2.6, we have $M_2^{(5)}\subseteq\mathcal{G}[[4,13]]$. It follows from $\mathcal{G}\setminus\{1\}$ is $M_3^{(5)}$-free that $\{2,3,14,15,16\}\notin\mathcal{G}$. This means that

$$\text{if } i\geq 2,\ \text{then } k\leq 13 \text{ for } ijk\ell m\in\mathcal{G}. \tag{2}$$

Thus,

$$\lambda(\mathcal{G})\leq f_{122},$$

where

$$\begin{aligned}f_{122}=&a\left(\binom{12}{4}\left(\frac{b}{12}\right)^4+\binom{12}{3}\left(\frac{b}{12}\right)^3 c+\binom{12}{2}\left(\frac{b}{12}\right)^2\frac{c^2}{2}+b\frac{c^3}{6}+\frac{c^4}{24}\right)\\&+\binom{12}{5}\left(\frac{b}{12}\right)^5+\binom{12}{4}\left(\frac{b}{12}\right)^4 c+\binom{12}{3}\left(\frac{b}{12}\right)^3\frac{c^2}{2}.\end{aligned}$$

Case 2: $M_3^{(5)}\subseteq\mathcal{G}\setminus\{1\}$.

By Lemma 2.6, it is easy to get $M_3^{(5)}\subseteq\mathcal{G}[[2,16]]$. Note that $\{1,17,18,19,20\}\notin\mathcal{G}$ and then

$$j\leq 16 \text{ for } ijk\ell m\in\mathcal{G}. \tag{3}$$ 2

Subcase 2.1: $M_3^{(5)}\subseteq\mathcal{G}\setminus\{1,2\}$.

By Corollary 2.6, it is easy to get $M_3^{(5)}\subseteq\mathcal{G}[[3,17]]$ and then $\{1,2,18,19,20\}\notin\mathcal{G}$. So,

$$k\leq 17 \text{ for } ijk\ell m\in\mathcal{G}. \tag{4}$$ 2.1

Subcase 2.1.1: $\mathcal{G}\setminus\{1,2,3,4\}$ is $M_2^{(5)}$-free.

In [2], it has been proved that $\pi_\lambda(M_2^{(5)})=5!\lambda(S_1^{(5)})$. So $\lambda(\mathcal{G}[[5,n]],\boldsymbol{x})\leq\lambda(S_1^{(5)})(b+c)^5\approx 3.41333\times 10^{-3}(b+c)^5$. Combining (3) and (4), we have

$$\lambda(\mathcal{G})\leq f_{211},$$

where

$$f_{211} = \binom{a}{4}^4 (b+c) + \binom{4}{3}\binom{a}{4}^3 \left(\binom{13}{2}\left(\frac{b}{13}\right)^2 + bc + \frac{c^2}{2} \right) + \binom{4}{2}\binom{a}{4}^2 \left[\binom{13}{3}\left(\frac{b}{13}\right)^3 + \binom{13}{2}\left(\frac{b}{13}\right)^2 c + b\frac{c^2}{2} \right] + a\left[\binom{13}{4}\left(\frac{b}{13}\right)^4 + \binom{13}{3}\left(\frac{b}{13}\right)^3 c + \binom{13}{2}\left(\frac{b}{13}\right)^2 \frac{c^2}{2} \right] + \lambda(S_1^{(5)})(b+c)^5.$$

Subcase 2.1.2: $M_2^{(5)} \subseteq \mathcal{G} \setminus \{1,2,3,4\}$.

Again, by Lemma 2.6, we get $M_2^{(5)} \subseteq \mathcal{G}[[5,14]]$ and then $\mathcal{G} \setminus \{1,2,3\}$ is $M_3^{(5)}$-free by Fact 2.2. So we have $\{4,15,16,17,18\} \notin \mathcal{G}$, which implies that

$$\text{if } i \geq 4, \text{ then } j \leq 14 \text{ for } ijk\ell m \in \mathcal{G}. \tag{5}$$ 2.1.2

If $\mathcal{G} \setminus \{1,2,3,4,5\}$ is $M_2^{(5)}$-free, then using the result in [2], we have $\lambda(\mathcal{G}[[6,n]], \boldsymbol{x}) \leq \lambda(S_1^{(5)})(c+d+e)^5 \approx 3.41333 \times 10^{-3}(c+d+e)^5$. Combining (3), (4) and (5), we have

$$\lambda(\mathcal{G}) \leq f_{212(a)},$$

where

$$f_{212(a)} = \binom{a}{3}^3 \left[\left(\frac{b}{2}\right)^2 + b(c+d+e) + X \right] + \binom{3}{2}\binom{a}{3}^2 \left[\left(\frac{b}{2}\right)^2 (c+d+e) + bX + Y \right] + a\left[\left(\frac{b}{2}\right)^2 X + bY + Z + \left(\frac{d}{3}\right)^3 e + \binom{3}{2}\left(\frac{d}{3}\right)^2 \frac{e^2}{2} \right] + \left(\frac{b}{2}\right)^2 Y + bZ + \lambda(S_1^{(5)})(c+d+e)^5$$

with

$$X = \binom{9}{2}\left(\frac{c}{9}\right)^2 + c(d+e) + \binom{3}{2}\left(\frac{d}{3}\right)^2 + de + \frac{e^2}{2},$$

$$Y = \binom{9}{3}\left(\frac{c}{9}\right)^3 + \binom{9}{2}\left(\frac{c}{9}\right)^2 (d+e) + c\left(\binom{3}{2}\left(\frac{d}{3}\right)^2 + de + \frac{e^2}{2} \right) + \left(\frac{d}{3}\right)^3 + \binom{3}{2}\left(\frac{d}{3}\right)^2 e + d\frac{e^2}{2},$$

and

$$Z = \binom{9}{4}\left(\frac{c}{9}\right)^4 + \binom{9}{3}\left(\frac{c}{9}\right)^3 (d+e) + \binom{9}{2}\left(\frac{c}{9}\right)^2 \left(\binom{3}{2}\left(\frac{d}{3}\right)^2 + de + \frac{e^2}{2} \right) + c\left(\left(\frac{d}{3}\right)^3 + \binom{3}{2}\left(\frac{d}{3}\right)^2 e + d\frac{e^2}{2} \right).$$

Otherwise, $M_2^{(5)} \subseteq \mathcal{G} \setminus \{1,2,3,4,5\}$. By Lemma 2.6, we have $M_2^{(5)} \subseteq \mathcal{G}[[6,15]]$. It follows from $\mathcal{G} \setminus \{1,2,3\}$ that $M_3^{(5)}$-free. So$\{4,5,16,17,18\} \notin \mathcal{G}$. Combining (3), (4) and (5), we have

$$\lambda(\mathcal{G}) \leq f_{212(b)},$$

where

$$f_{212(b)} = \binom{a}{3}^3 \left(\binom{12}{2} \left(\frac{b}{12}\right)^2 + bc + bd + \left(\frac{c}{2}\right)^2 + cd + \frac{d^2}{2} \right) + \binom{3}{2} \left(\frac{a}{3}\right)^2 X + aY$$
$$+ \binom{12}{5} \left(\frac{b}{12}\right)^5 + \binom{12}{4} \left(\frac{b}{12}\right)^4 (c+d) + \binom{12}{3} \left(\frac{b}{12}\right)^3 \left(\left(\frac{c}{2}\right)^2 + cd + \frac{d^2}{2} \right)$$

with

$$X = \binom{12}{3} \left(\frac{b}{12}\right)^3 + \binom{12}{2} \left(\frac{b}{12}\right)^2 (c+d) + b\left(\left(\frac{c}{2}\right)^2 + cd + \frac{d^2}{2} \right) + \left(\frac{c}{2}\right)^2 d + c\frac{d^2}{2},$$

and

$$Y = \binom{12}{4} \left(\frac{b}{12}\right)^4 + \binom{12}{3} \left(\frac{b}{12}\right)^3 (c+d) + \binom{12}{2} \left(\frac{b}{12}\right)^2 \left(\left(\frac{c}{2}\right)^2 + cd + \frac{d^2}{2} \right)$$
$$+ b\left(\left(\frac{c}{2}\right)^2 d + c\frac{d^2}{2} \right) + \left(\frac{c}{2}\right)^2 \frac{d^2}{2}.$$

Subcase 2.2: $\mathcal{G} \setminus \{1,2\}$ is $M_3^{(5)}$-free.

Subcase 2.2.1: $\mathcal{G} \setminus \{1,2,3\}$ is $M_2^{(5)}$-free.

If $\mathcal{G} \setminus \{1,2,3,4\}$ is $M_1^{(5)}$-free, then by (3), we have

$$\lambda(\mathcal{G}) \le f_{221(a)},$$

where

$$f_{221(a)} = \left(\frac{a}{4}\right)^4 (b+c) + \binom{4}{3} \left(\frac{a}{4}\right)^3 \left(\binom{12}{2} \left(\frac{b}{12}\right)^2 + bc + \frac{c^2}{2} \right)$$
$$+ \binom{4}{2} \left(\frac{a}{4}\right)^2 \left[\binom{12}{3} \left(\frac{b}{12}\right)^3 + \binom{12}{2} \left(\frac{b}{12}\right)^2 c + b\frac{c^2}{2} + \frac{c^3}{6} \right]$$
$$+ a\left[\binom{12}{4} \left(\frac{b}{12}\right)^4 + \binom{12}{3} \left(\frac{b}{12}\right)^3 c + \binom{12}{2} \left(\frac{b}{12}\right)^2 \frac{c^2}{2} + b\frac{c^3}{6} \right].$$

Otherwise, by Lemma 4.1, we have $\lambda(\mathcal{G} \setminus \{1,2\}) \le 3.66902 \times 10^{-3}$. Then by (3), we have

$$\lambda(\mathcal{G}) \le f_{221(b)},$$

where

$$f_{221(b)} = \left(\frac{a}{2}\right)^2 \left[\binom{14}{3} \left(\frac{b}{14}\right)^3 + \binom{14}{2} \left(\frac{b}{14}\right)^2 c + b\frac{c^2}{2} + \frac{c^3}{6} \right] + a\left[\binom{14}{4} \left(\frac{b}{14}\right)^4 \right.$$
$$\left. + \binom{14}{3} \left(\frac{b}{14}\right)^3 c + \binom{14}{2} \left(\frac{b}{14}\right)^2 \frac{c^2}{2} + b\frac{c^3}{6} \right] + \lambda(\mathcal{G} \setminus \{1,2\})(b+c)^5.$$

Subcase 2.2.2: $M_2^{(5)} \subseteq \mathcal{G} \setminus \{1,2,3\}$.

By Lemma 2.6, we have $M_2^{(5)} \subseteq \mathcal{G}[[4,13]]$. Since $\mathcal{G} \setminus \{1,2\}$ is $M_3^{(5)}$-free, $\{3,14,15,16,17\} \notin \mathcal{G}$. We have

$$\text{if } i \ge 3, \text{then } j \le 13 \text{ for } \ ijk\ell m \in \mathcal{G}. \tag{6}$$ 2.2.2

Subcase 2.2.2.1: $\mathcal{G}\setminus\{1,2,3,4\}$ is $M_2^{(5)}$-free.

If $\{5,6,12,13,14\}\notin\mathcal{G}$, then

$$\text{if } i\geq 5,\ \text{then } k\leq 11 \text{ for } ijk\ell m\in\mathcal{G}. \tag{7}$$ a

Combining (3), (6) and (7), we have

$$\lambda(\mathcal{G})\leq f_{2221(a)},$$

where

$$\begin{aligned} f_{2221(a)}=&\left(\frac{a}{2}\right)^2\left(\left(\frac{b}{2}\right)^2(c+d+e+f)+bX+Y\right)+a\left[\left(\frac{b}{2}\right)^2X+bY+Z\right.\\ &\left.+\left(\frac{e}{3}\right)^3f+\binom{3}{2}\left(\frac{e}{3}\right)^2\frac{f^2}{2}+e\frac{f^3}{6}\right]+\left(\frac{b}{2}\right)^2Y+bZ+\binom{7}{5}\left(\frac{c}{7}\right)^5\\ &+\binom{7}{4}\left(\frac{c}{7}\right)^4(d+e+f)+\binom{7}{3}\left(\frac{c}{7}\right)^3W \end{aligned}$$

with

$$W=\left(\frac{d}{2}\right)^2+d(e+f)+\binom{3}{2}\left(\frac{e}{3}\right)^2+ef+\frac{f^2}{2},$$

$$X=\binom{7}{2}\left(\frac{c}{7}\right)^2+c(d+e+f)+W,$$

$$\begin{aligned} Y=&\binom{7}{3}\left(\frac{c}{7}\right)^3+\binom{7}{2}\left(\frac{c}{7}\right)^2(d+e+f)+cW+\left(\frac{d}{2}\right)^2(e+f)\\ &+d\left(\binom{3}{2}\left(\frac{e}{3}\right)^2+ef+\frac{f^2}{2}\right)+\left(\frac{e}{3}\right)^3+\binom{3}{2}\left(\frac{e}{3}\right)^2f+e\frac{f^2}{2}+\frac{f^3}{6} \end{aligned}$$

and

$$\begin{aligned} Z=&\binom{7}{4}\left(\frac{c}{7}\right)^4+\binom{7}{3}\left(\frac{c}{7}\right)^3(d+e+f)+\binom{7}{2}\left(\frac{c}{7}\right)^2W\\ &+c\left(\left(\frac{d}{2}\right)^2(e+f)+d\left(\binom{3}{2}\left(\frac{e}{3}\right)^2+ef+\frac{f^2}{2}\right)+\left(\frac{e}{3}\right)^3+\binom{3}{2}\left(\frac{e}{3}\right)^2f+e\frac{f^2}{2}+\frac{f^3}{6}\right)\\ &+\left(\frac{d}{2}\right)^2\left(\binom{3}{2}\left(\frac{e}{3}\right)^2+ef+\frac{f^2}{2}\right)+d\left(\left(\frac{e}{3}\right)^3+\binom{3}{2}\left(\frac{e}{3}\right)^2f+e\frac{f^2}{2}+\frac{f^3}{6}\right). \end{aligned}$$

Otherwise, it follows from $\mathcal{G}\setminus\{1,2,3,4\}$ is $M_2^{(5)}$-free that $\{7,8,9,10,11\}\notin\mathcal{G}$, which means that $\mathcal{G}[[7,n]]=\emptyset$. In this case, if $\{5,10,11,12,13\}\in\mathcal{G}$, then $\{6,7,8,9,14\}\notin\mathcal{G}$ and so

$$\text{if } i\geq 6,\ \text{then } m\leq 13 \text{ for } ijk\ell m\in\mathcal{G}. \tag{8}$$ b1

Combining (3), (6) and (8), we have

$$\lambda(\mathcal{G})\leq f_{2221(b_1)},$$

where

$$f_{2221(b_1)} = \binom{a}{2}^2 \left[\binom{b}{3}^3 + \binom{3}{2}\binom{b}{3}^2 (c+d+e+f) + b\,(c(d+e+f)+X) + cX + Y \right]$$
$$+ a\left[\binom{b}{3}^3 (c+d+e+f) + \binom{3}{2}\binom{b}{3}^2 \Big(c(d+e+f)+X\Big) + b(cX+Y) \right.$$
$$\left. + cY + Z + \binom{e}{3}^3 f + \binom{3}{2}\binom{e}{3}^2 \frac{f^2}{2} + e\frac{f^3}{6} \right] + \binom{b}{3}^3 \Big(c(d+e+f)+X\Big)$$
$$+ \binom{3}{2}\binom{b}{3}^2 (cX+Y) + b(cY+Z) + c\binom{7}{4}\binom{d}{7}^4$$

with

$$X = \binom{7}{2}\binom{d}{7}^2 + d(e+f) + \binom{3}{2}\binom{e}{3}^2 + ef + \frac{f^2}{2},$$

$$Y = \binom{7}{3}\binom{d}{7}^3 + \binom{7}{2}\binom{d}{7}^2 (e+f) + d\left(\binom{3}{2}\binom{e}{3}^2 + ef + \frac{f^2}{2}\right)$$
$$+ \binom{e}{3}^3 + \binom{3}{2}\binom{e}{3}^2 f + e\frac{f^2}{2} + \frac{f^3}{6}$$

and

$$Z = \binom{7}{4}\binom{d}{7}^4 + \binom{7}{3}\binom{d}{7}^3 (e+f) + \binom{7}{2}\binom{d}{7}^2 \left(\binom{3}{2}\binom{e}{3}^2 + ef + \frac{f^2}{2}\right)$$
$$+ d\left(\binom{e}{3}^3 + \binom{3}{2}\binom{e}{3}^2 f + e\frac{f^2}{2} + \frac{f^3}{6}\right).$$

Otherwise,

$$\text{if } i \geq 5, \text{ then } j \leq 9 \text{ for } ijk\ell m \in \mathcal{G}. \tag{9}$$ b2

Combining (3), (6) and (9), we have

$$\lambda(\mathcal{G}) \leq f_{2221(b_2)},$$

where

$$f_{2221(b_2)} = \binom{a}{2}^2 \left[\binom{b}{2}^2 (c+d+e+f+g) + b\left(\binom{c}{2}^2 + c(d+e+f+g) + X\right) \right.$$
$$\left. + \binom{c}{2}^2 (d+e+f+g) + cX + Y \right] + a\left[\binom{b}{2}^2 \left(\binom{c}{2}^2 + c(d+e+f+g) + X\right) \right.$$
$$+ b\left(\binom{c}{2}^2 (d+e+f) + cX + Y\right) + \binom{c}{2}^2 X + cY + Z + W + \binom{f}{3}^3 g$$
$$\left. + \binom{3}{2}\binom{f}{3}^2 \frac{g^2}{2} + f\frac{g^3}{6} \right] + \binom{b}{2}^2 \left(\binom{c}{2}^2 (d+e+f+g) + cX + Y\right)$$
$$+ b\left(\binom{c}{2}^2 X + cY + Z + W\right) + \binom{c}{2}^2 Y + cZ$$

with

$$V = \binom{4}{3}\left(\frac{e}{4}\right)^3 + \binom{4}{2}\left(\frac{e}{4}\right)^2(f+g) + e\left(\binom{3}{2}\left(\frac{f}{3}\right)^2 + fg + \frac{g^2}{2}\right) + \left(\frac{f}{3}\right)^3 + \binom{3}{2}\left(\frac{f}{3}\right)^2 g + f\frac{g^2}{2} + \frac{g^3}{6},$$

$$W = \left(\frac{e}{4}\right)^4 + \binom{4}{3}\left(\frac{e}{4}\right)^3(f+g) + \binom{4}{2}\left(\frac{e}{4}\right)^2\left(\binom{3}{2}\left(\frac{f}{3}\right)^2 + fg + \frac{g^2}{2}\right) + e\left(\left(\frac{f}{3}\right)^3 + \binom{3}{2}\left(\frac{f}{3}\right)^2 g + f\frac{g^2}{2} + \frac{g^3}{6}\right),$$

$$X = \binom{3}{2}\left(\frac{d}{3}\right)^2 + d(e+f+g) + \binom{4}{2}\left(\frac{e}{4}\right)^2 + e(f+g) + \binom{3}{2}\left(\frac{f}{3}\right)^2 + fg + \frac{g^2}{2},$$

$$Y = \left(\frac{d}{3}\right)^3 + \binom{3}{2}\left(\frac{d}{3}\right)^2(e+f+g) + d\left(\binom{4}{2}\left(\frac{e}{4}\right)^2 + e(f+g) + \binom{3}{2}\left(\frac{f}{3}\right)^2 + fg + \frac{g^2}{2}\right) + V,$$

and

$$Z = \left(\frac{d}{3}\right)^3(e+f+g) + \binom{3}{2}\left(\frac{d}{3}\right)^2\left(\binom{4}{2}\left(\frac{e}{4}\right)^2 + e(f+g) + \binom{3}{2}\left(\frac{f}{3}\right)^2 + fg + \frac{g^2}{2}\right) + dV.$$

Subcase 2.2.2.2: $M_2^{(5)} \subseteq \mathcal{G} \setminus \{1,2,3,4\}$.

By Lemma 2.6, we have $M_2^{(5)} \subseteq \mathcal{G}[[5,14]]$. It follows from $\mathcal{G} \setminus \{1,2\}$ is $M_2^{(5)}$-free that $\{3,4,15,16,17\} \notin \mathcal{G}$. This means that

$$\text{if } i \geq 3, \text{ then } k \leq 14 \text{ for } ijk\ell m \in \mathcal{G}. \tag{10}$$ 2.2.2.2

Combining (3) and (10), we have

$$\lambda(\mathcal{G}) \leq f_{2222},$$

where

$$f_{2222} = \left(\frac{a}{2}\right)^2\left(\binom{12}{3}\left(\frac{b}{12}\right)^3 + \binom{12}{2}\left(\frac{b}{12}\right)^2(c+d) + bX + Y\right) + a\left[\binom{12}{4}\left(\frac{b}{12}\right)^4 + \binom{12}{3}\left(\frac{b}{12}\right)^3(c+d) + \binom{12}{2}\left(\frac{b}{12}\right)^2 X + bY + \left(\frac{c}{2}\right)^2\frac{d^2}{2} + c\frac{d^3}{6}\right] + \binom{12}{5}\left(\frac{b}{12}\right)^5 + \binom{12}{4}\left(\frac{b}{12}\right)^4(c+d) + \binom{12}{3}\left(\frac{b}{12}\right)^3 X,$$

with

$$X = \left(\frac{c}{2}\right)^2 + cd + \frac{d^2}{2} \text{ and } Y = \left(\frac{c}{2}\right)^2 d + c\frac{d^2}{2} + \frac{d^3}{6}.$$

All argument shows that $\lambda(\mathcal{G}) \leq \lambda\left(S_3^{(5)}\right)$, which completes the proof. ∎

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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